AMATH/PMATH 331: Applied Real Analysis Fall 2026

Assignment 4; due Tuesday, November 10, 2026

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[1] Let $A \subseteq \mathbb{R}^n$ be a non-empty subset, and let $\mathrm{Map}(A, \mathbb{R}^m)$ be the space of maps from A to $\mathbb{R}^m$.

[a] Let (f_k) be a sequence in $\mathrm{Map}(A, \mathbb{R}^m)$ which converges uniformly to a *bounded* function f, so $f \in B(A, \mathbb{R}^m)$. Prove that the f_k are *eventually bounded*. That is, prove that there exists $N \in \mathbb{N}$, such that $k \geq N$ implies $f_k \in B(A, \mathbb{R}^m)$.

[b] Suppose (f_k) and (g_k) are two sequences in $\mathrm{Map}(A, \mathbb{R}^m)$ which converge uniformly to *bounded* functions $f, g \in B(A, \mathbb{R}^m)$. Prove that the sequence $(\langle f_k, g_k \rangle)_{k \in \mathbb{N}}$ in $\mathrm{Map}(A, \mathbb{R})$ converges uniformly to $f \cdot g \in \mathrm{Map}(A, \mathbb{R})$, where $\langle \cdot, \cdot \rangle$ is the dot product in $\mathbb{R}^m$.

[c] Show that the assumption of boundedness of the limits in part [b] is essential as follows. Show that the sequence (f_k) in $M(\mathbb{R}, \mathbb{R})$ given by $f_k(x) = x + \frac{1}{k}$ converges uniformly on $\mathbb{R}$ to the unbounded function $f(x) = x$, but that $f_k^2 \in M(\mathbb{R}, \mathbb{R})$ does not converge uniformly to f^2.

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[2] Let $f \in \mathrm{Map}(\mathbb{R}^n, \mathbb{R}^m)$ be a map, and suppose that f is *continuous* at $\mathbf{x}_0 \in \mathbb{R}^n$.

Define a sequence (f_k) in $\mathrm{Map}(\mathbb{R}^n, \mathbb{R}^m)$ by $f_k(\mathbf{x}) = f(\mathbf{x}_0 + \frac{1}{k}\mathbf{x})$. Let $g \in \mathrm{Map}(\mathbb{R}^n, \mathbb{R}^m)$ be the *constant map* $g(\mathbf{x}) = f(\mathbf{x}_0)$ for all $\mathbf{x} \in \mathbb{R}^n$.

Prove that, for any $R > 0$, the sequence (f_k) converges uniformly to g on the set $B_R(\mathbf{0})$.

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[3] For $k \in \mathbb{N}$, define $f_k \colon [-1, 1] \to \mathbb{R}$ by $f_k(x) = (1 - x^2)^k$.

[a] For any $\delta \in (0, 1)$, prove that the sequence of functions (f_k) *converge uniformly* on the subset $[-1, -\delta] \cup [\delta, 1]$ of $\mathbb{R}$.

[b] Prove that the sequence of functions (f_k) *does not converge uniformly* on the subset $[-1, 1]$ of $\mathbb{R}$.

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[4] Let $A \subseteq \mathbb{R}^n$ be non-empty. Let (f_k) be a sequence in $\mathrm{Map}(A, \mathbb{R}^m)$ which converges *uniformly* to a limit $f \in \mathrm{Map}(A, \mathbb{R}^m)$. We proved in class that if every f_k is continuous on A, then f is also continuous on A.

[a] Adapt that proof to show that if every f_k is *uniformly* continuous on A, then f is also *uniformly* continuous on A.

[b] Explain why we do not need to do any extra work if A is compact. That is, why does part [a] follow immediately if A is compact?

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[5] Let $a < b$. Recall the Weierstrass Approximation Theorem: if $f \in \mathcal{C}[a, b]$, then there exists a sequence (p_k) in $\mathcal{P}[a, b]$ that converges uniformly to f on $[a, b]$.

Suppose that we have proved this theorem for the particular case $[a, b] = [0, 1]$. Show that the general case of any $[a, b]$ follows from that.

Hint: Consider the map $T \colon \mathbb{R} \to \mathbb{R}$ given by $T(x) = a + (b - a)x$.

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[6] Let $f \in \mathcal{C}[0, 1]$. Suppose that $\int_0^1 x^n f(x)\, dx = 0$ for all integers $n \geq 0$.

[a] Find a way to use the Weierstrass Approximation Theorem to show that $\int_0^1 |f(x)|^2\, dx = 0$.
Hint: Consider the integral of $f(f - p)$ for an appropriately chosen $p \in \mathcal{P}[0, 1]$.

[b] Explain why this means that $f(x) = 0$ for all $x \in [0, 1]$.
Hint: Suppose that there exists some $a \in [0, 1]$ where $f(a) \neq 0$, and try to derive a contradiction using continuity of f and properties of integration.

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[7] Suppose that $f\colon [0,1] \to \mathbb{R}$ is *continuously differentiable*. This means that:

- f is differentiable at all points in $[0,1]$, where we define differentiability at the end points using one-sided limits, *and*
- the derivative $f'\colon [0,1] \to \mathbb{R}$ is continuous on $[0,1]$, so $f' \in \mathcal{C}[0,1]$.

Show that there exists a sequence of polynomials (p_k) in $\mathcal{P}[0,1]$ such that (p_k) converges uniformly to f *and* the sequence (p_k') of derivatives converges uniformly to f'.

Hint: Start with f', and consider the fundamental theorem of calculus.

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[8] Let $a > 0$ and $c \in \mathbb{R}$. Consider the map $T\colon \mathbb{R} \to \mathbb{R}$ given by $T(x) = ax^2 + c$.

[a] Show that, depending on a, c, the map T may have two, one, or zero fixed points.

[b] Show that if there is only one fixed point, we cannot classify it as attracting or repelling.

[c] Show that if there are two fixed points, one is always repelling, and the behaviour of the other depends on whether ac is greater or less than $-\frac{3}{4}$, and we learn nothing if $ac = -\frac{3}{4}$.

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[9] Let $\lambda > 0$. Consider the map $T\colon \mathbb{R} \to \mathbb{R}$ given by $T(x) = x + \frac{\lambda}{\pi}\sin(\pi x)$.

[a] Compute all of the fixed points of T. You should find that there are infinitely many, and they are very simple to describe.

[b] Show that "half" of the fixed points are always *repelling*.

[c] Find a condition on λ which ensures that the other "half" of the fixed points are *attracting*, and find a different condition on λ which ensures that this other half is repelling.

[d] For what value of λ do we know nothing about this other half of the fixed points?

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[10] Let $\lambda > 0$. Consider the map $T\colon \mathbb{R} \to \mathbb{R}$ given by $T(x) = x + \frac{\lambda}{2\pi}\sin^2(\pi x)$.

[a] Compute all of the fixed points of T. You should find that there are infinitely many, and they are very simple to describe.

[b] What can you say qualitatively about the fixed points in this case?
